

FINANCE CANON

2026-03-18

Dexter Hadley, MD/PhD

Hadley Lab CANONIC

Abstract

INDUSTRY is the variable in the SERVICE equation every vertical wires primitives into a governed domain.

hadleylab.org Governed Research. Every claim cited.

Contents

0.1	1. Auditability	1
0.2	2. Regulatory Compliance	1
0.3	3. Risk Governance	2
0.4	4. Medical Innovation Finance	2
0.5	5. Digital Asset Compliance	2
0.6	6. Insurance & Actuarial	2
0.7	7. Tax Governance	3

1 Constraints 3

- **INDUSTRY_IS_VARIABLE** SERVICE equals PRIMITIVE composed with INDUSTRY; swap the industry, get a new service.
- **EVERY_INDUSTRY_WIRES_THREE** each vertical wires INTEL, CHAT, and COIN before it counts as governed.
- **LANGUAGE_CASCADES_FROM_MAGIC** industry language inherits from MAGIC; no per-industry DESIGN or lexicon fork. ## Axioms

0.1 1. Auditability

Financial claims **MUST** be auditable. Every transaction, balance, and disclosure requires an independent verification trail.

Example: US GAAP (ASC 606 revenue recognition, ASC 842 leases) vs IFRS (IFRS 15, IFRS 16) two global frameworks, each requiring complete audit trails. SOX Section 404 requires management assessment and external auditor attestation of internal controls over financial reporting. PCAOB (Public Company Accounting Oversight Board) sets auditing standards for public companies. Material weakness in internal controls triggers accelerated disclosure (8-K). Audit trail = every journal entry traceable to source document.

0.2 2. Regulatory Compliance

Financial operations **MUST** satisfy applicable regulatory requirements across all jurisdictions.

Example: SEC (Securities Act of 1933, Securities Exchange Act of 1934) registration, disclosure, anti-fraud provisions. FINRA (broker-dealer regulation) licensing (Series 7, 63, 65), suitability/Reg BI (best interest), margin rules. OCC regulates national banks. CFPB (Consumer Financial Protection Bureau) enforces consumer protection (TILA, ECOA, FCRA). FinCEN BSA/AML compliance: CTRs (\$10K+), SARs (suspicious activity), CDD (Customer Due Diligence) rule, beneficial ownership require-

ments. Dodd-Frank Act (2010) created CFPB, Volcker Rule (proprietary trading ban), orderly liquidation authority, derivatives regulation (Title VII).

0.3 3. Risk Governance

Risk MUST be quantified, monitored, and disclosed. Capital adequacy MUST meet regulatory minimums.

Example: Basel III/IV minimum capital requirements (CET1 4.5%, Tier 1 6%, Total Capital 8%), leverage ratio (3%), liquidity (LCR, NSFR), countercyclical buffer. Dodd-Frank stress testing (DFAST) and CCAR (Comprehensive Capital Analysis and Review) annual Fed assessment of largest banks. VaR (Value at Risk), Expected Shortfall (ES), and scenario analysis. CECL (Current Expected Credit Loss, ASC 326) forward-looking loss provisioning. Enterprise risk management: COSO framework, three lines of defense model.

0.4 4. Medical Innovation Finance

Healthcare innovation MUST be financed through regulatory-aware capital structures.

Example: FDA approval pathway economics: 510(k) submission (\$13,320 fee, 3-6 month review, ~\$31M average total development cost) vs PMA (\$425,000+ fee, 12-18 month review, ~\$94M average total cost). CMS reimbursement coding: CPT codes for procedures, HCPCS Level II for devices/supplies, ICD-10-PCS for inpatient procedures. New Technology Add-on Payment (NTAP) provides incremental reimbursement for innovative devices. Medical device tax (2.3%, currently suspended). Orphan Drug Act 7-year market exclusivity, 25% tax credit on clinical trial costs, FDA fee waivers. SBIR/STTR grants: Phase I (~\$275K/6 months),

Phase II (~\$1M/2 years) non-dilutive funding for small businesses including medtech.

0.5 5. Digital Asset Compliance

Digital assets MUST be classified and regulated according to their function and jurisdiction.

Example: SEC Howey test (SEC v. W.J. Howey Co., 1946) investment of money in a common enterprise with expectation of profits from others efforts = security. CFTC classifies certain tokens as commodities (Bitcoin, Ether under certain circumstances). State money transmitter licensing 49 states + DC each have separate requirements (BitLicense in New York). MiCA (Markets in Crypto-Assets, EU 2024) comprehensive framework for crypto-asset service providers. Travel Rule (FATF) VASPs must share originator/beneficiary information for transfers above thresholds.

0.6 6. Insurance & Actuarial

Insurance operations MUST comply with state-based regulatory frameworks and actuarial standards.

Example: McCarran-Ferguson Act (1945) insurance regulation is primarily state, not federal. NAIC (National Association of Insurance Commissioners) develops model laws adopted by states. State insurance departments regulate rates, forms, solvency, and market conduct. ACA (Affordable Care Act) medical loss ratio: 80% for individual/small group, 85% for large group insurers must spend this percentage of premiums on claims and quality improvement. Reinsurance treaties and retrocession. Actuarial Standards of Practice (ASOPs) govern pricing, reserving, and risk assessment. RBC (Risk-Based Capital) formula determines minimum capital requirements.

0.7 7. Tax Governance

Tax obligations MUST be computed, documented, and filed according to applicable tax law.

Example: IRC (Internal Revenue Code) corporate tax rate 21% (TCJA 2017). R&D tax credit (Section 41) 20% of qualified research expenses above base amount, or 14% simplified method. QSBS exclusion (Section 1202) up to 100% exclusion on gain from qualified small business stock held 5+ years (up to \$10M or 10x basis). Transfer pricing (Section 482, OECD Guidelines) arms-length standard for intercompany transactions. Section 174 (post-2022) mandatory capitalization and amortization of R&D expenses (5 years domestic, 15 years foreign). Opportunity Zones (Section 1400Z) tax incentives for investment in designated census tracts.

1. Constraints

MUST: Cite specific regulation or standard for financial claims

MUST: Distinguish US GAAP from IFRS where applicable

MUST NOT: Present financial projections without risk disclosure framework

FINANCE / CANON / VERTICALS